

Calculus II (MATH 2205)
Fall 2017

Name (Print): _____

Quiz 10: Section 9.3, 9.4

Score: _____ / 10

Section 9.3

1. 5 points Find the first four nonzero terms of the Maclaurin series for the function

$$f(x) = (1 + x^2)^{-1}.$$

Then, write the Maclaurin series for the function using summation notation and determine the interval of convergence of the series.

Section 9.4

2. 5 points Evaluate the following limit using Taylor series.

$$\lim_{x \rightarrow 0} \frac{1 + x - e^{-x}}{4x^2}.$$